

Sri Akash Kadali

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EDUCATION

University of Maryland, College Park, United States

CGPA: 3.67/4

Master of Science in Applied Machine Learning

August 2024 - May 2026

SKILLS

Languages: C++

Databases: Database operations, Testing and validating database operations, Testing and Validating

Tools: Debugging, Documentation, Documentation of results

Other: 3D geometry, 3D software, Imaging hardware/software, Imaging hardware and software, Medical imaging, Geometry algorithms, 3D Geometry Problems, Quality control (QC), QC/defects, Quality Control, Testing/validation

EXPERIENCE

Machine Learning Engineer Intern

May 2025 – August 2025

Ayar Labs

Santa Clara, CA

- Built C++ + SQL test harnesses that test and validate database operations, verifying expected read/write behavior for imaging hardware/software workflows and executing 127 automated read/write scenarios with deterministic pass/fail results.
- Validated 3D geometry problem-solving tasks by implementing and verifying geometry computations in C++ to align optical transceiver test measurements, confirming numerical correctness across 3D constraint sets and documenting results for review.
- Diagnosed validation failures by identifying and assigning defects for proper QC tracking, reproducing database and 3D processing issues from logs, then filing 19 QC/defects with evidence and reproduction steps.

Software Engineer Intern

July 2024 – December 2024

Indian Institute of Technology, Indore

Indore, India

- Designed and ran repeatable test cases to exercise database operations under realistic conditions, integrating SQL test automation with imaging data ingestion patterns.
- Quantified 3D geometry output correctness by comparing computed results to expected constraints using C++ math checks, improving numerical consistency across 50+ validation scenarios and flagging edge-case regressions earlier.
- Profiled validation failures by debugging and profiling software behavior and system interactions through execution traces, surfacing 46 previously untracked data gaps and pinpointing root causes to improve defect assignment accuracy.

Software Engineer Intern

January 2024 – June 2024

National Institute of Technology, Jaipur

Jaipur, India

- Accelerated C++ development by implementing and iterating on C++ code changes to support database validation and 3D geometry problem solutions, stabilizing computations across edge cases.
- Facilitated technical documentation of testing and validation results, including evidence and step-by-step repros for regression defects, reducing triage time by 7% and aligning 39 cross-functional stakeholders across 4 teams.
- Analyzed QC workflows by ensuring defects in a bug tracking system were properly categorized and linked to validation evidence, improving traceability; scanned 74M rows in 1 second using 6 validation rules.

Software Engineer Intern

May 2023 – December 2023

Indian Institute of Technology, Indore

Indore, India

- Partnered with senior 3D software engineers to solve challenging 3D software engineering problems through code reviews and joint debugging, reducing reproduction time for geometry-related failures from 3 days to 18 hours.
- Engaged in a fast-paced environment by reprioritizing tasks and adapting to shifting engineering needs, executing agile planning with Jira task tracking to unblock 14 downstream teams.
- Contributed by presenting technical progress and results to the team through brief updates and/or structured summaries, including documentation of results from imaging hardware/software integration tests, defect assignments, and quality outcomes.

PROJECTS

C++ 3D Geometry Validation Harness with Vector-Indexed QC for Imaging Metadata

- Tools used:** C++, SQLite, SQLite3, GoogleTest, Eigen, FAISS
- Built a C++ test harness that solved advanced 3D geometry constraints and persisted imaging metadata via validated database read/write flows.
- Detected 31% more geometry/QC defects by enforcing deterministic invariants and automated DB consistency checks across 50K randomized cases.

ACHIEVEMENTS

Published "CaDT-Net: Cascaded Deformable Transformer for Breast Cancer" at ICONIP 2024, achieving 99% accuracy in image classification.

Solved 200+ LeetCode problems, focusing on Graphs, DP, and System Design.